

Bagging

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Recap

- Last time we covered **Tree Methods II**:
 - Regression trees.
 - Procedure.
 - Interpreting results.
 - Predictions.

Introduction

- Tree methods are easy to construct and relatively easy to interpret.
- In the presence of high variability, they can struggle to generalize to new data.
- Combining multiple models (ensembles) can help with generalizing to new data.

Recall: Bootstrapping

- **Bootstrapping:**

- If the sample is large enough (looks like the population), we can pretend that the sample is actually the population.
- We then take samples from our original sample **with replacement** of the same size as the original sample.
- The resulting distribution is not the true sampling distribution, but is an approximation called the **bootstrap distribution**.

Recall: Bootstrapping Procedure

- 1 Randomly select an observation from the original sample, which was drawn from the population.
- 2 Record the observation's value.
- 3 Replace that observation.
- 4 Repeat steps 1–3 (sampling with replacement) until you have n observations, which form a bootstrap sample.
- 5 Calculate the bootstrap point estimate (e.g., mean, median, proportion, slope, etc.) of the n observations in your bootstrap sample.
- 6 Repeat steps 1–5 many times to create a distribution of point estimates (the bootstrap distribution).
- 7 Calculate the plausible range of values around our observed point estimate.

Recall: Bootstrapping Procedure

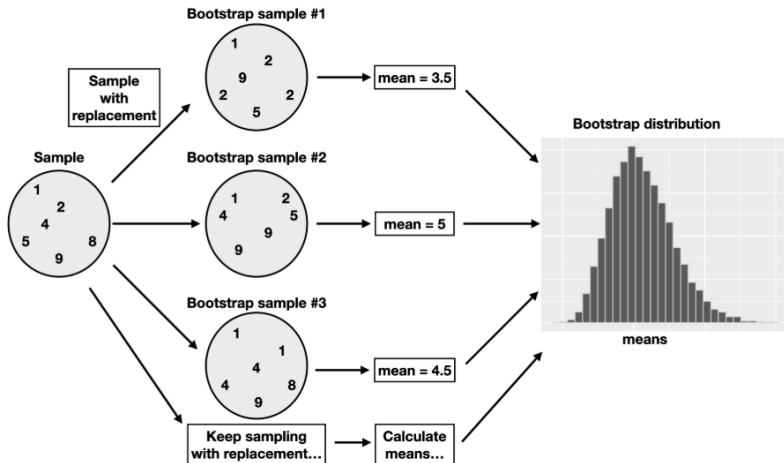


Figure: Source: (3)

Recall: Bootstrapping in R

- *Many functions have the option to estimate using bootstrapped samples.*
- We can simply use the `rep_sample_n(size = n, replace = TRUE, reps = r)` function from *infer*.
 - You can also use the functionality of *dplyr*.
- 95% bootstrap C.I. in R:
 - ```
bounds <- bootstrapped.values %>%
 dplyr::select(values)%>%
 pull()%>%
 quantile(c(0.025, 0.975))
```
- **Remember to set your seed for reproducibility!**



## Example 1

- 1 Set your seed to 3040 and take a random sample of  $n = 500$  observations from a lognormal distribution  $\text{Lognormal}(\mu = 50, \sigma = 0.5)$ .
- 2 Generate a histogram of your sample.
  - What is your sample median?
- 3 Take 1000 bootstrapped samples from your original sample.
- 4 Generate a histogram of the bootstrapped distribution of the median.
- 5 What is the 95% bootstrapped confidence interval for the population median?

# Introduction to Ensemble Methods

- **Ensemble methods** combine the the outputs of *base* models to improve prediction accuracy.
- There are two primary types of ensemble techniques:
  - ① **Bagging** (Bootstrap aggregation): Generates different training datasets through bootstrapping and training a separate model on each of these datasets.
  - ② **Boosting**: Models trained sequentially, where each model attempts to correct the errors of the previous ones.

## Bootstrap Aggregating

## Bootstrap Aggregating

- **Bootstrap aggregating (bagging)** fits multiple versions of a prediction model and then combines them into an aggregated prediction.
  - Models are fit from bootstrapped samples of the original sample.
- It is especially useful in situations with high variance.
- *Not as effective in cases of low variability or high bias.*

## Illustrative Example 1

- 1 Run the provided code.
- 2 Compare the differences in applying bagging to polynomial regression and regression trees.
- 3 What do you notice?

## Bagging Mechanism

- Procedure:
  - ①  $b$  bootstrap copies of the original training data are created.
  - ② The regression/classification algorithm is applied to each bootstrap sample.
  - ③ Predictions are made from each of the *base learners*.
  - ④ New predictions are made by averaging the predictions together from the individual base learners.
- *Classification problems: the base learner predictions are combined using plurality vote or by averaging the estimated class probabilities together.*

## Illustrative Example 2

- 1 Run the provided code to generate the animation.
- 2 What do you notice?

## Bagging in R

- We can use the `bagging()` function from the *ipred* package in R.
  - In Python

- Usage:

```
model <- bagging(
 formula = response ~ X1+X2+...+XP,
 data = df,
 nbagg = b,
 coob = TRUE,
 control = rpart.control(minsplit = 2, cp = 0)
)
```



## Example 2

- Load the *ptitanic* data from the *rpart* package and do the following:
  - ① Set your seed and use the provided code to split your data into a 75/25 train and test set.
  - ② Construct an out-of-the-box classification tree using the training set with `survived` as the response.
  - ③ Construct bootstrap aggregated tree with 160 trees.
  - ④ Compare the accuracies on the testing set.
- What do you notice?

## Cross-Validation

- We can also use the *caret* package to perform cross-validation on our bagged trees:

```
model <- train(
 response ~ X1+X2+...+XP,
 data = train,
 method = "treebag",
 trControl = trainControl(method = "cv", number = 10),
 nbagg = b,
 control = rpart.control(minsplit = 2, cp = 0)
)
```

## Example 3

- Load the *ptitanic* data from the *rpart* package and adjust the code do the following:
  - 1 Set your seed.
  - 2 Perform cross-validation on an out-of-the-box classification tree using the training set with `survived` as the response.
  - 3 Perform cross-validation on bootstrap aggregated trees with 160 trees each.
- What do you notice?

## Illustrative Example 3

- **Running everything sequentially can be very computationally demanding.**
- Import the Wages.csv data and do the following:
  - 1 Split your data into a training and testing set.
  - 2 Run the code to set up parallel clusters to speed up the process.
  - 3 Perform parallel bagging and get predictions from 160 decision trees.
  - 4 Generate the plot to help to determine how many bagged trees are needed for the best RMSE.
  - 5 Remember to shutdown the parallel cluster.
- What did you notice?

## Number of Trees

- As your tasks become more complex, averaging many trees may become computationally demanding.
- You can use some accuracy measure to help select the number of trees required.
  - *Eventually, this measure should plateau.*
- **Remember, these averages are still biased by your original sample.**

## Variable Importance

## Variable Importance

- It can be difficult to determine the *significance* of variables.
- May measure variable importance based on the sum of the reduction in the loss function of each variable at each split in a given tree.
- Using the vip and caret packages in R:
  - `vip(caret.model)`

## Example 4

- Use your results from Example 3 to determine the importance of the variables in the *ptitanic* data.
- What do you notice?



## Final Thoughts

- Bagging improves the prediction accuracy for high variance models at the expense of interpretability and computational speed.
- **The trees are not completely independent of each other.**
  - Run Illustrative Example 4 in the code.
- **Tree correlation** prevents bagging from further reducing the variance of the base learner.
  - Random forests can help with this.

## Next Lecture

- We will discuss **Random Forests**.
- Procedure.
- Tree complexity and tuning.

## Exercise 1

- Import the wine dataset from the *HDclassif* package.
- Create a 75-25 train/test split.
- Construct an out-of-the box classification tree for your training dataset.
- Find the ideal complexity parameter (cp) using cross validation and construct this tree.
- **Use what we have covered in these slides to construct Bootstrap Aggregated trees.**
- Finally, compare the classification accuracy of all of your trees on the testing set.
- What do you notice?

## Exercise 2

- Import the diamonds dataset from the *ggplot2* package.
- Create a 75-25 train/test split.
- Construct an out-of-the box regression tree for the price of the diamonds in your training dataset.
- Find the ideal complexity parameter (cp) using cross validation and construct this tree.
- **Use what we have covered in these slides to construct Bootstrap Aggregated trees.**
- Finally, compare the accuracy of all of your trees on the testing set.

## References & Resources

- 1 Bakker, J. D. (2026) Applied Multivariate Statistics in R
  - 2 Peng R. D. (2020) *Exploratory Data Analysis with R*
  - 3 Timbers T., Campbell T., and Lee M. (2024) *Data science: A first introduction*. Chapman and Hall/CRC
  - 4 Bradley Boehmke B., and Greenwell B. (2020) *Hands-On Machine Learning with R*. Taylor & Francis
- bagging()
  - rpart()
  - rpart.plot()
  - vip()
  - foreach
  - doParallel